

Retail Sales & Profitability Analysis

End-to-End Analysis of the Superstore Dataset using Python, SQL Server, and Power BI

1. Project Overview

This project analyzes four years of retail transaction data (2014–2017) from a national Superstore chain to understand where the business makes and loses money. The goal was to answer a concrete business brief — identify unprofitable product lines, quantify the impact of discounting on margin, and evaluate customer, regional, and operational performance — and deliver the findings as a decision-ready dashboard, mirroring how an analytics request would move through a real company: from raw data, to a governed data model, to a stakeholder-facing report.

2. Dataset Summary

- Rows: 9,994 order line items
- Columns: 21 (raw) — expanded to 24 after feature engineering
- **Date Range:** January 3, 2014 – December 30, 2017
- **Key Features:**
 - Order details (Order ID, Order/Ship Date, Ship Mode)
 - Customer details (Customer ID, Name, Segment)
 - Product details (Product ID, Name, Category, Sub-Category)
 - Geography (City, State, Postal Code, Region)
 - Transaction metrics (Sales, Quantity, Discount, Profit)
- **Data Quality:** 0 missing values, 0 duplicate rows across all 9,994 records — confirmed via `isnull()` and `duplicated()` checks.

3. Exploratory Data Analysis using Python

Data preparation and validation were carried out in Python (pandas) before the data was modeled in SQL:

- **Data Loading:** Imported the raw .xls export using `pandas.read_excel()`.
- **Initial Exploration:** Used `df.info()`, `df.describe(include='all')`, and `df.isnull().sum()` to confirm structure, types, and completeness.
- **Column Standardization:** Renamed columns to snake_case (lower-case, spaces/hyphens replaced with underscores) for consistency across Python, SQL, and Power BI.
- **Data Integrity Checks:** Verified zero duplicate rows and zero null values; confirmed no `ship_date` occurred before its `order_date`.
- **Feature Engineering:**
 - `profit_margin` — profit as a percentage of sales, per order line
 - `shipping_days` — (`ship_date` – `order_date`), used for fulfillment analysis
 - `order_year`, `order_month`, `order_year_month` — date parts for trend analysis
 - `is_loss` — boolean flag marking any order with negative profit
- **Outlier Analysis:** Isolated the bottom 1% of orders by profit (100 rows) and found every single one carried a discount of 40% or higher — the earliest signal that discount depth, not order size, was driving losses.

- **Database Integration:** Loaded the cleaned DataFrame into Microsoft SQL Server via SQLAlchemy/pyodbc (df.to_sql()) for structured analysis and star-schema modeling.

Key statistic validated in Python before moving to SQL:

Discount Level	Order Count	Profit Margin
Below 40% discount	8,855	19.93%
40% discount or more	1,139	-50.04%

4. Data Modeling: Star Schema (SQL Server)

Rather than analyzing one flat table, the cleaned data was split into a star schema in SQL Server — a single Fact table holding transaction-level measures, and four Dimension tables holding descriptive attributes. This removes redundant text data, keeps the model in sync with a single source of truth, and mirrors how BI teams structure data for reporting.

Table	Type	Key Columns
Fact_Orders	Fact	order_id, customer_id, product_id, geo_id, order_date, sales, quantity, discount, profit
Dim_Customer	Dimension	customer_id, customer_name, segment
Dim_Product	Dimension	product_id, product_name, category, sub_category
Dim_Geography	Dimension	geo_id, postal_code, city, state, region
Dim_Date	Dimension	date, year, month, quarter, year_month (built in Power BI)

Fact_Orders joins to Dim_Geography on postal_code, city, state, and region to resolve each order's location key, keeping only IDs and numeric measures in the fact table. Dim_Product and Dim_Customer were deduplicated using GROUP BY / ROW_NUMBER() to guarantee one row per product_id and customer_id — required for valid one-to-many relationships in Power BI.

5. Data Analysis using SQL (Business Transactions)

Eight business queries were written against the star schema to answer the questions in the brief, each mirroring a real ad-hoc request an analyst might get from a stakeholder:

1. Monthly Sales & Profit Trend

Aggregated sales and profit by order_month to reveal seasonal patterns across the 4-year window.

2. Top Sub-Categories by Profit Margin

Ranked all sub-categories by profit ÷ sales to find the most efficient product lines.

3. Sub-Categories That Lose Money

Filtered to sub-categories with negative total profit. Result:

Sub-Category	Total Profit
Tables	-\$17,725.48
Bookcases	-\$3,472.56

Supplies	-\$1,189.10
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4. Discount Impact on Profit

Bucketed every order into discount bands (0%, 1–20%, 21–40%, 40%+) and summed profit per band. Profit declines steadily as discount depth increases, collapsing entirely once discount reaches 40% or more — confirming the pattern first seen in the Python outlier check.

5. Top 10 Customers by Lifetime Sales

Ranked customers by cumulative sales. Sean Miller (\$25,043), Tamara Chand (\$19,052), and Raymond Buch (\$15,117) lead the list.

6. Year-over-Year Growth by Region

Sales aggregated by region and order_year to track regional growth trajectories across 2014–2017.

7. Customer Recency (RFM-style)

Identified customers with no order in the trailing 12 months relative to the dataset's latest date — a churn-risk flag.

8. Shipping Performance by Mode

Average fulfillment time per ship_mode:

Ship Mode	Avg. Shipping Days
Same Day	0.04
First Class	2.18
Second Class	3.24
Standard Class	5.01

6. Dashboard in Power BI

The final deliverable is a 4-page interactive Power BI dashboard connected to the SQL Server star schema, built with DAX time-intelligence and ranking measures rather than static visuals.

Data Model

- Fact_Orders related to all four dimension tables (one-to-many, single-direction cross-filter)
- Dim_Date marked as an official Date Table to support time-intelligence functions
- Core DAX measures: Total_Sales, Total_Profit, Profit_Margin%, Total_Orders, Sales_YoY%, Sales_MTD/YTD, Discount Band, Shipping Days, Is New Customer

Page 1 — Executive Summary

- KPI cards: Total Sales, Total Profit, Profit Margin %, Sales YoY %
- Sales & Profit trend line by month/year
- Sales-by-state filled map
- Year and Region slicers, plus a callout strip surfacing the top-performing state and overall growth rate

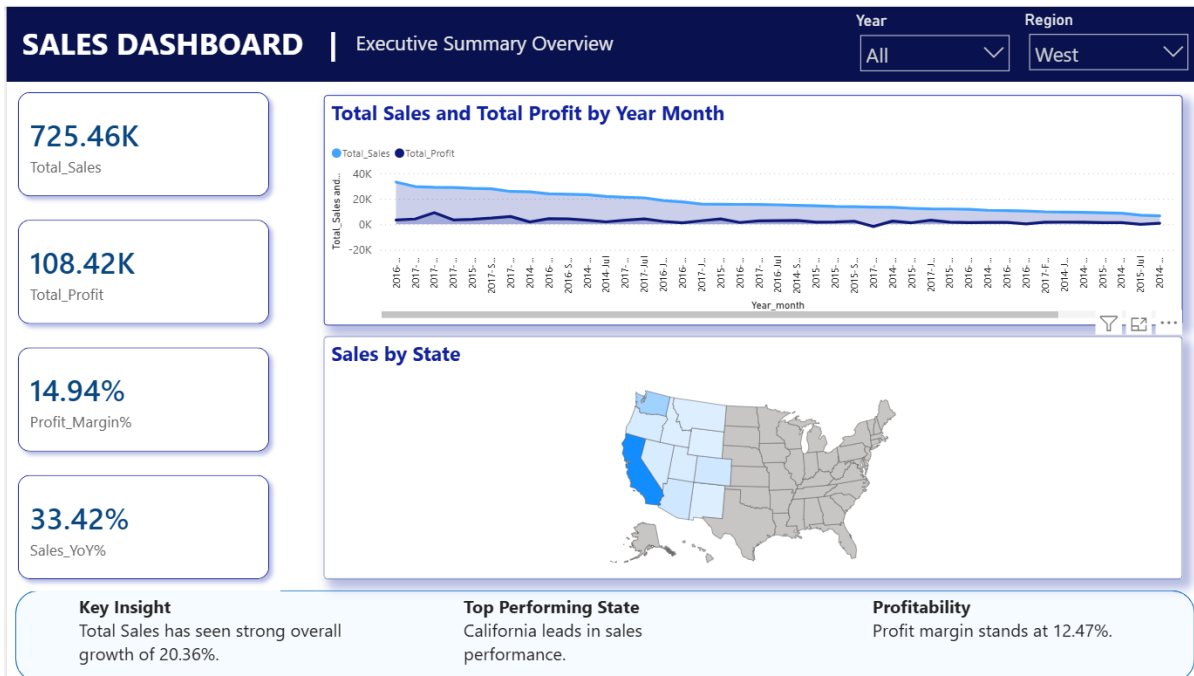


Figure 1: Executive Summary — shown filtered to the West region (KPIs reflect \$725.46K sales, \$108.42K profit, 14.94% margin for that region only)

Page 2 — Sales & Profitability

- Sub-category profit chart with red/green conditional formatting — visually isolates Tables, Bookcases, and Supplies as the only loss-making lines
- Profit by Discount Band — shows the steady decline in profit as discount depth increases
- Top 10 and Bottom 10 products by profit (RANKX-based ranking measures)
- Year and Sub-Category slicers



Figure 2: Sales & Profitability — sub-category profit, discount-band impact, and top/bottom 10 products by total sales

Page 3 — Customer Insights

- Top 10 customers table by lifetime sales and profit
- Segment breakdown (Consumer / Corporate / Home Office) by sales
- New vs. Returning customer split, based on each customer's first order year
- Year and Segment slicers

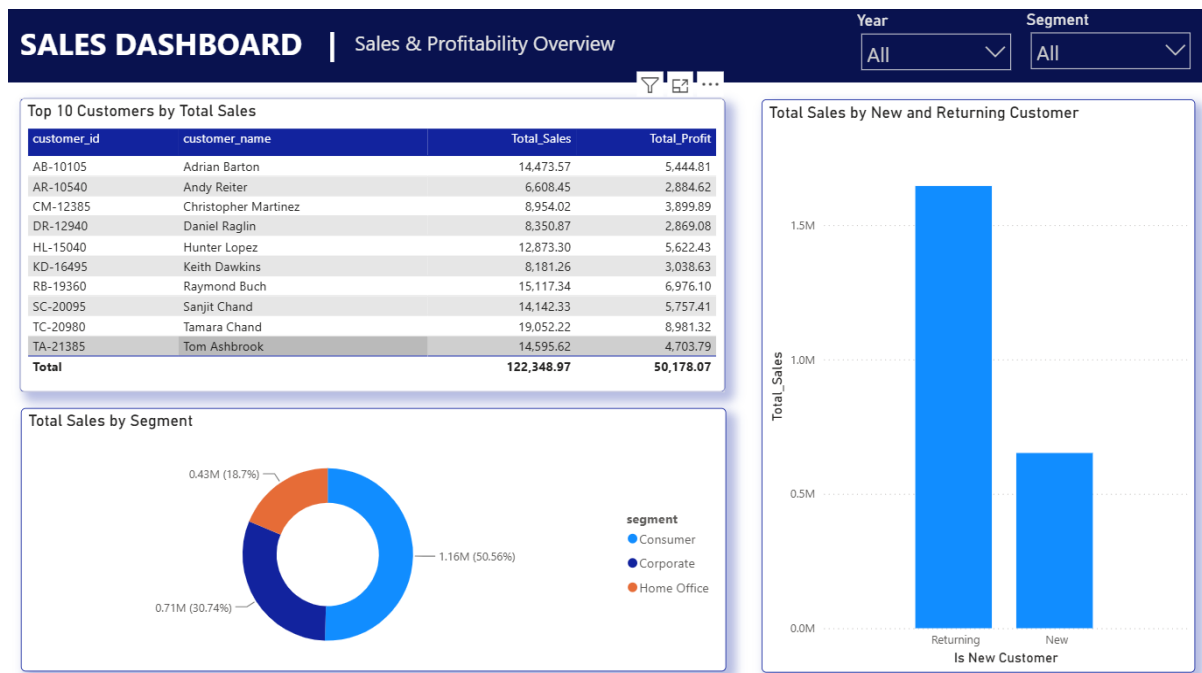


Figure 3: Customer Insights — top 10 customers, segment mix, and new vs. returning customer sales

Page 4 — Operations

- Average shipping days by ship mode
- Late shipment rate and average fulfillment time KPI cards
- Orders by month split by ship mode, and shipping-days distribution across modes
- Year and Ship Mode slicers

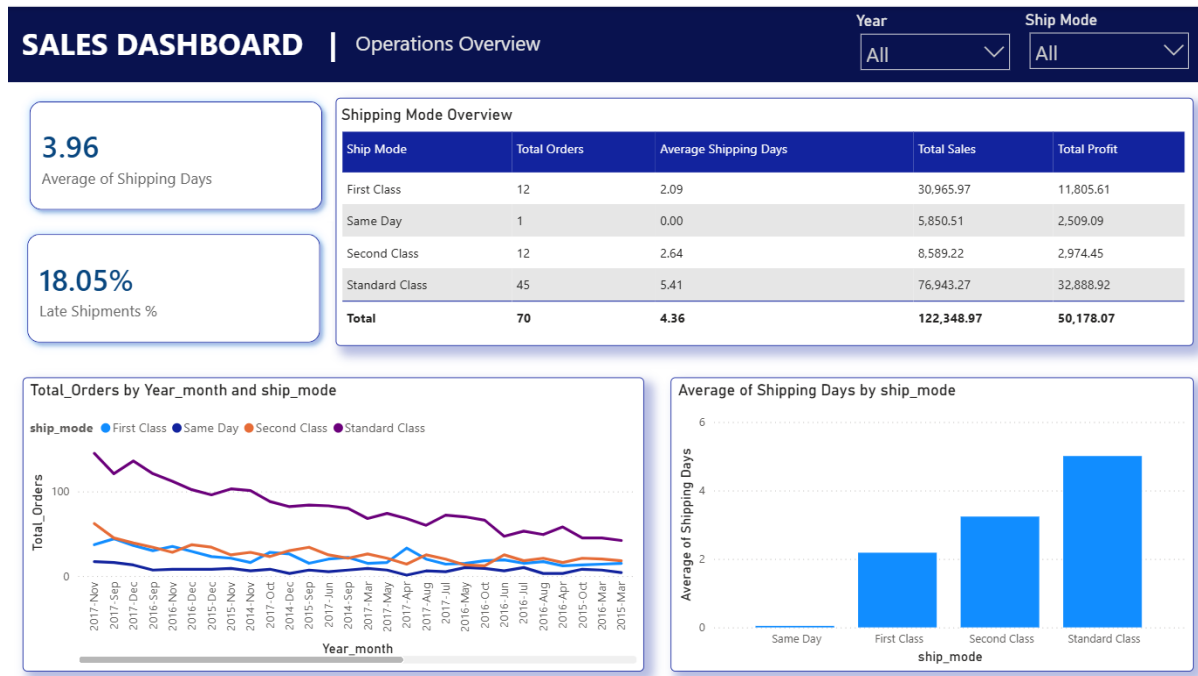


Figure 4: Operations — 3.96-day average fulfillment time, 18.05% late-shipment rate, and shipping performance by mode

7. Key Insights

- **Discounting above 40% is unprofitable at scale:** orders with a discount of 40% or more post an average profit margin of -50.04% , versus $+19.93\%$ for orders discounted below 40% — despite representing only 11% of total orders.
- **Losses are concentrated in three sub-categories:** Tables ($-\$17,725$), Bookcases ($-\$3,473$), and Supplies ($-\$1,189$) are the only sub-categories that lose money overall, out of 17 — all three carry an outsized share of deep-discount orders.
- **The West and East regions drive the most profit:** $\$108,418$ and $\$91,523$ respectively, more than double the Central region's $\$39,706$, despite Central having comparable order volume.
- **Consumer is the largest and most profitable segment:** $\$134,119$ in profit on $\$1.16\text{M}$ in sales, ahead of Corporate ($\$91,979$) and Home Office ($\$60,299$).
- **Shipping speed varies sharply by mode:** Same Day orders ship same-day on average (0.04 days) as expected, while Standard Class — the most common mode — averages 5.01 days.
- **Overall business health is solid:** $\$2.30\text{M}$ in total sales generated $\$286\text{K}$ in profit, a 12.47% blended margin, before accounting for the drag from deep-discount orders.

8. Business Recommendations

- **Cap discounts on loss-making sub-categories:** Restrict discounting above 30–40% on Tables, Bookcases, and Supplies, or require managerial approval past that threshold.
- **Reassess Tables pricing/sourcing:** As the single largest loss driver, Tables merits a margin review independent of the discount policy — cost structure or list pricing may also be a factor.
- **Reinforce investment in the West and East regions:** Both regions outperform on profit relative to sales volume and are strong candidates for expanded inventory or marketing spend.
- **Prioritize the Consumer segment:** As the highest-profit segment, targeted retention campaigns here likely carry the best return.

- **Investigate Standard Class delays:** With an average 5-day fulfillment time and the highest order volume, Standard Class is the biggest lever for improving overall customer experience.